Impact of AMC-Oriented CW Adversarial Attacks on CRC Integrity

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ABSTRACT

This paper studies a practical two-stage receiver pipeline in which automatic modulation classification (AMC) controls downstream demodulation and CRC verification. Using the existing `crc_experiment.py` implementation, we run a two-track evaluation: Track A uses real RML2016.10a IQ data to measure AWN behavior under clean and Carlini-Wagner (CW) attack conditions; Track B uses synthetic bursts with known payload bits to measure CRC pass rates. Perturbations are generated on Track A and transferred to Track B. Across digital modulations, average AMC accuracy drops from 97.19% to 4.65% under CW. However, CRC under oracle demodulation changes only from 63.62% to 59.59%. CRC collapse appears when demodulation is driven by attacked AMC output: Adv+AMC CRC falls to 4.12%. At 18 dB, the gap is stronger: 93.25% (Adv+Oracle) vs. 7.81% (Adv+AMC). The results indicate that the dominant failure path is AMC misclassification leading to wrong demodulator selection, not only direct waveform destruction.

KEYWORDS

adversarial machine learning, modulation classification, RF security, CRC, CW attack

1 INTRODUCTION

Practical receivers often use a two-stage pipeline: automatic modulation classification (AMC) followed by modulation-specific demodulation and CRC checking. This architecture is effective under noise, but it introduces a strong dependency: if AMC is manipulated, the receiver may select the wrong demodulator, and CRC can fail even when the waveform is still demodulatable under the correct modulation assumption.

This paper analyzes that failure path using the implementation in `crc_experiment.py` from the current codebase. We focus on three questions:

(1) Does CW mainly break AMC, waveform demodulability, or both?
(2) How large is the gap between oracle demodulation and AMC-driven demodulation under attack?
(3) Are the conclusions stable across low and high SNR regimes (0 and 18 dB)?

2 METHOD

2.1 Two-Track Design

The experiment uses a two-track setup.

(1) **Track A (AMC):** real `RML2016.10a` IQ samples are classified by AWN on clean and CW-attacked inputs to obtain AMC accuracy and perturbations.
(2) **Track B (CRC):** synthetic bursts with known payload bits are generated, demodulated, and validated with CRC pass/fail outcomes.

Code references: `crc_experiment.py:73`, `crc_experiment.py:175`, `crc_experiment.py:222`.

2.2 Transfer Perturbation

CW perturbations are generated on Track A and then transferred to Track B:

$$x_{\text{adv,synth}} = x_{\text{synth}} + \delta_{\text{real}}, \tag{1}$$

where $\delta_{\text{real}} = x_{\text{adv,real}} - x_{\text{real}}$.

Code references: `crc_experiment.py:212`, `crc_experiment.py:238`.

2.3 Evaluation Scenarios

For each modulation-SNR cell, four scenarios are evaluated:

(1) **Clean+Oracle:** clean IQ, true modulation used for demodulation.
(2) **Clean+AMC:** clean IQ, demodulation chosen from AMC behavior.
(3) **Adv+Oracle:** adversarial IQ, true modulation used for demodulation.
(4) **Adv+AMC:** adversarial IQ, demodulation chosen from attacked AMC output.

The adversarial window is spliced back into full burst context before demodulation to preserve guard-related filtering behavior (`crc_experiment.py:93–124`).

3 EXPERIMENTAL SETUP

3.1 Model and Data

(1) AMC model: AWN loaded from `2016.10a_AWN.pkl`.
(2) Real data (Track A): `./data/RML2016.10a_dict.pkl`.
(3) Synthetic data (Track B): generated by `util.synth_txrx.generate_burst`
(4) CRC-capable digital modulations: BPSK, QPSK, 8PSK, QAM16, QAM64, PAM4, CPFSK, GFSK.

3.2 Attack and Output Scope

(1) Attack: CW via `create_attack('cw', ...)`.
(2) Track A uses 500 real samples per modulation-SNR cell.
(3) Result file for this paper: `crc_experiment_results/crc_vs_amc.csv`.
(4) Track B count in this result: $n = 200$ synthetic bursts per digital cell.
(5) SNR settings in the result file: 0 and 18 dB.

Table 1: Overall Means on Digital Modulations

Scenario	AMC Acc (%)	CRC Pass (%)
Clean+Oracle	97.19	63.62
Clean+AMC	97.19	62.12
Adv+Oracle	4.65	59.59
Adv+AMC	4.65	4.12

Table 2: SNR-Stratified Means on Digital Modulations

Scenario (SNR)	AMC Acc (%)	CRC Pass (%)
Clean+Oracle (0 dB)	96.28	27.81
Clean+AMC (0 dB)	96.28	26.94
Adv+Oracle (0 dB)	2.30	25.94
Adv+AMC (0 dB)	2.30	0.44
Clean+Oracle (18 dB)	98.10	99.44
Clean+AMC (18 dB)	98.10	97.31
Adv+Oracle (18 dB)	7.00	93.25
Adv+AMC (18 dB)	7.00	7.81

Table 3: CRC Pass Rate at 18 dB by Modulation

Modulation	Clean+Oracle	Adv+Oracle	Adv+AMC
BPSK	100.0	100.0	39.0
QPSK	100.0	100.0	1.5
8PSK	100.0	100.0	0.5
QAM16	100.0	99.5	0.5
QAM64	96.0	76.0	0.5
PAM4	100.0	71.0	2.0
CPFSK	100.0	100.0	18.5
GFSK	99.5	99.5	0.0

4 RESULTS

4.1 Overall Means on Digital Modulations

Table 1 reports means over all digital cells (8 modulations × 2 SNR levels × 4 scenarios).

CW collapses AMC (97.19% → 4.65%). However, with oracle demodulation, CRC drops only moderately (63.62% → 59.59%). CRC collapse occurs primarily in Adv+AMC.

4.2 SNR-Stratified Means

At 18 dB, the gap between Adv+Oracle and Adv+AMC CRC is 85.44 points (93.25% vs. 7.81%), highlighting the AMC-induced error path.

4.3 Per-Modulation CRC at 18 dB

In 13 out of 16 digital cells, the Clean+Oracle and Adv+Oracle CRC difference is within 2 percentage points. In contrast, 14 out of 16 cells have Adv+AMC CRC at or below 5%.

5 DISCUSSION

5.1 Primary Failure Mechanism

The dominant path is not always direct waveform destruction. Instead, the main mechanism is: CW causes AMC misclassification, then the receiver selects the wrong demodulator, leading to CRC failure.

5.2 Why 18 dB Is More Diagnostic

At 0 dB, several modulations already have low clean CRC baselines, which mixes channel-limited errors with attack-induced errors. At 18 dB, clean baselines are near saturation, so attack effects are easier to isolate.

5.3 Modulation-Specific Sensitivity

QAM64 and PAM4 show notable Adv+Oracle drops (96.0% to 76.0%, and 100.0% to 71.0%, respectively), suggesting direct sensitivity beyond classification errors.

6 THREATS TO VALIDITY

(1) The study is transfer-based: perturbations are generated on Track A real data and applied to Track B synthetic bursts. Conclusions are about transferability, not full real link attacks.
(2) Clean+AMC and Adv+AMC in Track B are distribution-driven simulations based on Track A AMC behavior, not per-sample AMC inference on synthetic bursts.
(3) The CSV artifact does not fully encode all CW hyperparameters used at runtime unless captured in command logs.
(4) Analog modulations (WBFM, AM-DSB, AM-SSB) are reported without CRC metrics ($n = 0$).

7 CONCLUSION

Under the evaluated setup, CW nearly disables AMC, and CRC collapse is mainly realized through the AMC-dependent demodulation path. This identifies a practical system risk: classification robustness is not an isolated objective but a direct reliability dependency for link integrity.

ARTIFACT AND REPRODUCIBILITY

All numeric results in this paper come from `crc_experiment_results/crc_vs_an`
A baseline reproduction command is:

```
python crc_experiment.py --ckpt_path ./checkpoint \
  --output_dir ./crc_experiment_results
```

For exact reproducibility, explicitly set and record `-n_bursts`, `-cw_c`, `-cw_steps`, `-seed`, and `-ta_box`.

REFERENCES

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